LOSS Function that MINIMIZES Overfitting

Let $\underline{Y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}_{N_{\times 1}}$

Original Labels

from Training Samples

Let $\hat{Y} = \begin{pmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_n \end{pmatrix}_{N_{\times 1}}$

Predicted Labels

from Model $\hat{Y} = \underline{\mathbf{X}}\ \underline{\mathbf{W}}$

Objective is $\underline{\hat{Y}=Y}$ and $||W||$ is minimum This is called **Ridge Regularization**

Objective function $\mathbf{J(W)}=\mathbf{Loss}(\underline{X},\underline{Y},\underline{W})+\lambda\mathbf{R}(\underline{\mathbf{W}})$ (λ is a weighting factor)

Define **Loss** function as **squared distance** between $\hat{\boldsymbol{Y}}$ and $\boldsymbol{Y}$

Loss Term = $||Y-\hat{Y}||^2 = (Y-\hat{Y})^T(Y-\hat{Y})$

Regularization Term $R = ||W||^2 = W^TW$

Objective function $\mathbf{J(W)}$ $J(W) = \frac{1}{2}[(Y-\hat{Y})^\top (Y-\hat{Y}) + \lambda W^\top W]$